

TITLE PAGE

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- **Class Code:** [BIT1123]
- **Program:** Bachelor of Computer Science (Hons) (Software Engineering)
- **NRIC/Passport Number:** [P12371102]
- **Assignment Title:** Assignment 1 - Individual

1. Introduction

This self-reflective report documents my learning journey through the BIT1123 Object Oriented Programming course. The objective of the tutorials was to practically apply theoretical OOP concepts using Java, preparing me for real-world software engineering scenarios.

2. Knowledge Gained from Each Tutorial

Throughout the 10 weeks, I steadily built my programming knowledge:

- **Weeks 1-2:** I learned how to set up GitHub Codespaces, create basic Java classes, and instantiate objects.
- **Weeks 3-4 & 6:** I explored Inheritance and Polymorphism, learning how subclasses (Student, Lecturer) inherit from parent classes (Person, Employee).
- **Week 5:** I understood Encapsulation by protecting data using private variables and controlling access via getters and setters.
- **Week 7:** I practically applied Abstraction by hiding implementation details for smart home appliances.
- **Weeks 8-10:** I learned to use ArrayLists for dynamic data storage and implemented File I/O to read and write to text files persistently.

3. Challenges Encountered

The transition from simple procedural coding to complex OOP structures was initially challenging. One major difficulty was managing directory paths in the terminal, which often led to "File not found" errors when trying to compile Java files. Additionally, grasping the exact differences between Abstraction and regular Inheritance took some time.

4. How the Challenges Were Overcome

To overcome the terminal issues, I learned how to properly navigate directories using commands like `cd` and utilized the VS Code integrated terminal directly from the target folders. For conceptual misunderstandings, I engaged in repetitive practice, writing the code from scratch without looking at the templates until the logic of Abstraction and Encapsulation clicked.

5. Improvements in Java Programming Skills

My syntax memorization and debugging speed have vastly improved. I am now comfortable creating modular applications spanning multiple `.java` files rather than writing all code in a single file, which aligns with industry standards.

6. Understanding of Object-Oriented Programming Concepts

I now fully appreciate why OOP is a fundamental paradigm. It allows developers to model real-world entities efficiently. Concepts like Polymorphism allow code to be flexible, while Encapsulation ensures data security—both of which are critical for my future career in software development.

7. Future Learning Plans

Moving forward, I plan to dive deeper into Java frameworks like Spring Boot and explore Graphical User Interfaces (GUI) using JavaFX to upgrade my console-based applications into fully interactive software.

8. Conclusion

Building this GitHub repository from Week 1 to 10 has been highly rewarding. It serves as a personal portfolio showcasing my progression from a beginner to a capable Java programmer, providing a strong stepping stone for my academic and professional future.

9. GitHub Repository URL

https://github.com/pzqmm/Rashid-Mohamed-Omer_202501010714